1.07 Algebra: Simultaneous Equations

Worked Solutions

1 a)

$x = 1, y = 1$

1 b)

$x = 1, y = -\frac{1}{2}$

1 c)

$x - 9 + 3y = 0 \Rightarrow x + 3y = 9$

$2x + 2y = 2$

$x = -3, y = 4$

1 d)

$\frac{y}{2} + \frac{x}{4} = 5 \Rightarrow 2y + x = 20$

$x = 4, y = 8$

2 a)

$15 = 2x - 10 \Rightarrow x = \frac{25}{2}$

2 b)

$15 - \frac{25}{2}k = 10 \Rightarrow k = \frac{2}{5}$

3 a)

$\frac{y}{b} + \frac{11x}{80} = 1 \Rightarrow \frac{y}{b} = -\frac{11}{80}x + 1$

$\Rightarrow y = b(-\frac{11}{80}x + 1)$

$\Rightarrow y = -\frac{11}{80}bx + b$

$\therefore b = 8$

3 b)

$y = -\frac{11}{10}x + 8$

Never intersect $\Rightarrow$ same gradient

$a = \frac{11}{10}$

4)

$5 = 12 \times -\frac{1}{4} + c \Rightarrow c = 8$

$y = -\frac{1}{4}x + 8$

$5 = 12m + 1 \Rightarrow m = \frac{1}{3}$

$y = \frac{1}{3}x + 1$

$\therefore 4y + x = 32$ and $3y - x = 3$

5 a)

$2x + 4 = x^2 + 1$

$\Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x-3)(x+1) = 0$

$x = 3, x = -1$

$(3, 10), (-1, 2)$

5 b)

$3x - 2 = x^2 - 4x - 10$

$\Rightarrow x^2 - 7x - 8 = 0 \Rightarrow (x-8)(x+1) = 0$

$x = -1, x = 8$

$(-1, -5), (8, 22)$

5 c)

$\frac{1}{2}x = x^2 + 2x(\frac{1}{2}x)$

$\Rightarrow 0 = 2x^2 - \frac{1}{2}x \Rightarrow 0 = 4x^2 - x = x(4x-1)$

$x = 0, x = \frac{1}{4}$

$(0, 0), (\frac{1}{4}, \frac{1}{8})$

6 a)

$3y = x - 10 \Rightarrow y = \frac{1}{3}x - \frac{10}{3}$

$\Rightarrow x^2 + (\frac{1}{3}x - \frac{10}{3})^2 = 10$

$\Rightarrow x^2 + \frac{1}{9}x^2 - \frac{20}{9}x + \frac{100}{9} = 10$

$\Rightarrow \frac{10}{9}x^2 - \frac{20}{9}x + \frac{100}{9} = 10 \Rightarrow x^2 - 2x + 10 = 9$

$\Rightarrow x^2 - 2x + 1 = 0$ as req.

6 b)

$x^2 - 2x + 1 = 0 \Rightarrow (x-1)^2 = 0$

$x = 1$

$y = \frac{1}{3} \times 1 - \frac{10}{3} = -3$

$(1, -3)$

7 a)

$y = 15 - 3x \Rightarrow (15 - 3x)^2 = 9x - 27$

$\Rightarrow 9x^2 - 90x + 225 = 9x - 27$

$\Rightarrow 9x^2 - 90x + 225 = 9x - 27$

$\Rightarrow 9x^2 - 99x + 252 = 0$

$\Rightarrow x = 4, x = 7$

when $x = 4, y = 15 - 12 = 3$

when $x = 7, y = 15 - 21 = -6$

$(4, 3), (7, -6)$

7 b)

$y = 4 + x \Rightarrow x^2 + (4 + x) = 10$

$\Rightarrow x^2 + x - 6 = 0 \Rightarrow (x + 3)(x - 2) = 0$

$x = -3, x = 2$

When $x = -3, y = 1$

When $x = 2, y = 6$

$(-3, 1), (2, 6)$

7 c)

$y = 2x + 12 \Rightarrow x^2 + 2x(2x + 12) = 5$

$\Rightarrow x^2 + 4x^2 + 24x - 5 = 0 \Rightarrow 5x^2 + 24x - 5 = 0$

$\Rightarrow (5x - 1)(x + 5) = 0$

$x = \frac{1}{5}, x = -5$

When $x = \frac{1}{5}, y = 2 \times \frac{1}{5} + 12 = \frac{62}{5}$

When $x = -5, y = 2 \times -5 + 12 = 2$

$(\frac{1}{5}, \frac{62}{5}), (-5, 2)$

8 a)

$y = 4 - x \Rightarrow x^2 + (4 - x)^2 = 16$

$x^2 + 16 - 8x + x^2 = 16$

$\Rightarrow 2x^2 - 8x = 0 \Rightarrow 2x(x - 4) = 0$

$\Rightarrow x = 0, x = 4$

When $x = 0, y = 4$

When $x = 4, y = 0$

$(0, 4), (4, 0)$

8 b)

$y = 6 - 4x \Rightarrow x^2 + 2x + (6 - 4x)^2 - 3(6 - 4x) = 1$

$\Rightarrow x^2 + 2x + 36 - 48x + 16x^2 - 18 + 12x = 1$

$\Rightarrow 17x^2 - 34x + 17 = 0 \Rightarrow x = 1$

When $x = 1, y = 6 - 4 = 2$

$(1, 2)$

8 c)

$x = 8 - 2y \Rightarrow (8 - 2y)^2 + y^2 - (8 - 2y)y = 12$

$\Rightarrow 64 - 32y + 4y^2 + y^2 - 8y + 2y^2 = 12$

$\Rightarrow 7y^2 - 40y + 52 = 0$

$y = 2, y = \frac{26}{7}$

When $y = 2, x = 8 - 4 = 4$

When $y = \frac{26}{7}, x = 8 - \frac{52}{7} = \frac{4}{7}$

$(4, 2), (\frac{4}{7}, \frac{26}{7})$

8 d)

$y^2 + 8 = 3(2y + 1)y \Rightarrow y^2 + 8 = 6y^2 + 3y$

$\Rightarrow 5y^2 + 3y - 8 = 0$

$y = 1, y = -\frac{8}{5}$

When $y = 1, x = 2 \times 1 + 1 = 3$

When $y = -\frac{8}{5}, x = 2 \times -\frac{8}{5} + 1 = -\frac{11}{5}$

$(3, 1), (-\frac{11}{5}, -\frac{8}{5})$

9)

$x = 2y + 12 \Rightarrow (2y + 12)^2 + 2(2y + 12) + y^2 = 24$

$\Rightarrow 4y^2 + 48y + 144 + 4y + 24 + y^2 = 24$

$\Rightarrow 5y^2 + 52y + 144 = 0$

$y = \frac{-52 \pm \sqrt{2704 - 2880}}{10}$ which has no real solutions.

10)

$(5, 36)$ gives $36 = 25p - 5q + 1$

$(-2, 15)$ gives $15 = 4p + 2q + 1$

$35 = 25p - 5q$ and $14 = 4p + 2q$

$p = 2, q = 3$

11)

$(1, 6)$ gives $6 = p - 5 + 1 + q$

$(-\frac{1}{2}, \frac{21}{4})$ gives $\frac{21}{4} = -\frac{1}{8}p - 5 \times \frac{1}{4} - \frac{1}{2} + q$

$10 = p + q$ and $7 = -\frac{1}{8}p + q$

$p = \frac{8}{3}, q = \frac{22}{3}$

12)

$xy = 20$ and $x - y = \frac{8}{3}$

$\Rightarrow x = y + \frac{8}{3} \Rightarrow (y + \frac{8}{3})y = 20 \Rightarrow y^2 + \frac{8}{3}y - 20 = 0$

$y = \frac{10}{3}, y = -6$

When $y = \frac{10}{3}, x = 6$ (must be positive)

So $\frac{10}{3}$ and 6

13)

$2^{x-1} = (2^2)^{2y} \Rightarrow 2^{x-1} = 2^{4y} \Rightarrow x - 1 = 4y$

$9^{x-2} = 3^{y+1} \Rightarrow (3^2)^{x-2} = 3^{y+1} \Rightarrow 3^{2x-4} = 3^{y+1}$

$\Rightarrow 2x - 4 = y + 1$

So $x - 4y = 1$ and $2x - y = 5$

$x = \frac{19}{7}, y = \frac{3}{7}$

14)

$5^{2x-3} = (5^2)^{y-1} \Rightarrow 2x + 3 = 2y - 2$

$(2^4)^{x-1} = 2^{3y} \Rightarrow 4x - 4 = 3y$

So $2x - 2y = -5$ and $4x - 3y = 4$

$x = \frac{5}{2}, y = 2$